

Darin Tsui

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Education

Georgia Institute of Technology , Ph.D. in Electrical and Computer Engineering	Aug 2023 – May 2028
• Advisor: Amirali Aghazadeh	
University of California San Diego , B.S. in Bioengineering	Sep 2019 – Jun 2023

Selected Publications

Darin Tsui, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Advances in Neural Information Processing Systems* (NeurIPS) (Accepted), 2025.

Thomas A. Walton, **Darin Tsui**, Aryan Musharaf, and Amirali Aghazadeh. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *NeurIPS* (Accepted, **Spotlight**), 2025.

Darin Tsui, Kunal Talreja, and Amirali Aghazadeh. Sparse autoencoders for low- N protein function prediction and design. In *NeurIPS AI4Science Workshop* (Accepted), 2025.

Honors & Awards

NSF Graduate Research Fellowship	2024 - 2027
• Research proposal title: <i>Disease Variant Prediction Using Deep Generative Models</i> .	
Course Enhancements & Improvement (CREATION) Award , Georgia Tech	2023
• Award received for developing the course ECE 8803— <i>Generative and Geometric Deep Learning</i> .	
Graduate Student Conference Scholarship Award , Conference on Information Storage and Processing Systems (ISPS)	2023
President's Fellowship , Georgia Tech	2023
Anushka Michailova Memorial Bioengineering Award , UC San Diego	2023
• Award received for best capstone project out of 100 teams. Submissions were evaluated on technical depth, innovation, and research dissemination.	

Research Experience

AI ML and Information Research Group , Georgia Tech – Atlanta, GA	Jun 2023 – Present
• Advisor: Amirali Aghazadeh	
• Explained models 1000-fold quicker than state-of-the-art methods by designing algorithms at the intersection of <i>signal processing</i> , <i>coding theory</i> , <i>combinatorics</i> , and <i>interpretability</i> .	
• Leveraged <i>mechanistic interpretability</i> methods, such as sparse autoencoders and cross-layer transcoders, for extracting higher-order interactions, predicting protein function, and designing novel protein variants in the biosciences.	
• Designed and developed principled <i>generative models</i> that leverage domain priors.	
Integrated Systems Neuroengineering Lab , UC San Diego – San Diego, CA	Jun 2022 - Jun 2023
• Advisor: Gert Cauwenberghs. Mentored by Ph.D. candidates Akshay Paul, Tyler Bodily, and Min Lee.	
• Designed and developed COVID-19 diagnostic test using a graphene field-effect-transistor (GFET).	
• Integrated feature extraction and machine learning pipeline for current-voltage signals.	
• Achieved accuracies of 98.5% when detecting COVID-19 proteins, improving classification by 30.1%. Presented and published work at the <i>2023 IEEE Engineering and Medicine and Biology Conference (EMBC)</i> .	
Talke Biomedical Device Lab , UC San Diego – San Diego, CA	Dec 2021 - Aug 2023
• Advisor: Frank E. Talke	

- Designed low-cost vision system for minimally invasive surgery using OpenCV with fiducial markers.
- Integrated the tracking of ArUco markers on surgical platforms using Python and MATLAB.
- Implemented Kalman filtering to achieve sub-millimeter error in design-validation testing. Presented and published work in the journal *Microsystem Technologies* and the conferences *ISPS 2023* and *EMBC 2023*.

Swartz Center for Computational Neuroscience, UC San Diego – San Diego, CA

Dec 2021 - Aug 2023

- Advisor: Tzyy-Ping Jung. Mentored by Ph.D. candidate Chiyuan Chang.
- Designed electroencephalography-based (EEG) classification experiments. Conducted machine learning analysis of classroom stress levels.
- Performed support vector machine (SVM) and linear discriminant analysis (LDA) and Mann Whitney U statistical analysis using MATLAB and Python.
- Improved classification of stress vs. non-stress groups by 42.1% in correlation with physiological features.

Paesani Research Group, UC San Diego – San Diego, CA

Jan 2020 - Sep 2021

- Advisor: Francesco Paesani. Mentored by postdoctoral scholar Vinícius Cruzeiro.
- Designed and ran AMBER 20 molecular dynamics simulations in Bash and Python on supercomputing clusters to model amino acid and water molecule behaviors.
- Validated interactions using radial distribution function (RDF) fits.

Publications

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Thomas A. Walton, **Darin Tsui**, Lauren Fogel, Dustin J. E. Huard. Rafael S. Chagas, Raquel L. Lieberman, and Amirali Aghazadeh. GOLF: A generative AI framework for pathogenicity prediction of myocilin OLF variants. In *Machine Learning in Computational Biology*, 2025.

Darin Tsui, Kunal Talreja, and Amirali Aghazadeh. Efficient algorithm for sparse Fourier transform of generalized q -ary functions. In *IEEE Information Theory Workshop (ITW)*, 2025.

Darin Tsui and Amirali Aghazadeh. On recovering higher-order interactions from protein language models. In *International Conference on Learning Representations (ICLR) Generative and Experimental Perspectives for Biomolecular Design (GEM) Workshop*, 2024.

Teaching Experience

Teaching Assistant, Georgia Tech – Atlanta, GA

Aug 2023 - Dec 2023

- Advisor: Jeffery Hurley
- Graduate teaching assistant for ECE 4122/6122—*Advanced Programming Techniques for Engineering Applications*—with an enrollment of 300 graduate and undergraduate students. The course teaches undergraduate and graduate students techniques for distributed and parallel computing to run computational heavy tasks.
- Taught and assisted students in developing C++ code for lab assignments. Held office hours for 10 hours a week and mentored students on their final projects.

Instructional Assistant, UC San Diego – San Diego, CA

Dec 2020 - Mar 2021

- Advisor: Robert Sah
- Oversaw two lab sections of 30 students each for BENG 1—*Introduction to Bioengineering*.
- Taught lab sections covering topics ranging from biomechanics, noninvasive cardiovascular sensing, and spine segmentation. Held office hours and lab sections for 10 hours a week.

Industry Experience

Data Engineering Intern, Surgalign – San Diego, CA

Jan 2023 - Aug 2023

- Manager: Weidong Luo
- Interned at Surgalign, a small medical device company focused on developing AI solutions in spinal surgery (now acquired by Xtant Medical and Augmetics).
- Designed an internal MRI image processing application using Python that reduced preprocessing time.
- Assisted in developing ground truth dataset of imaging data and validated deep learning models against surgeon-generated data using Sørensen-Dice coefficient statistical testing.

Instruments R&D Intern, Johnson & Johnson – Redwood City, CA

Dec 2020 - Mar 2021

- Manager: Timothy Reppert
- Interned at Auris Health, Johnson & Johnson's flexible robotics division for minimally invasive surgery of urology. Supported design validation clinical trials through preparation and sterilization of instruments.
- Performed root cause analysis on the failure of instrument devices. Designed near-infrared camera system for calibration in endoscopes.
- Supported systems investigations towards optical calibration stations and instrumentation platforms.

Presentations

Darin Tsui. Explainability and mechanistic interpretability in AI, *presented as a guest lecturer for ECE 8803—Generative and Geometric Deep Learning—at Georgia Tech*, Atlanta, GA, 2025.

Darin Tsui. Generative and geometric deep learning in computational biology, *presented as a guest lecturer for ECE 8803—Generative and Geometric Deep Learning—at Georgia Tech*, Atlanta, GA, 2023.

Darin Tsui. Surgical site localization with non-invasive skin markers for pain management, *presented at the Center for Memory and Recording Research (CMRR) Research Review*, San Diego, CA, 2023.

Darin Tsui. Bioelectronic COVID-19 detection via graphene-field-effect-transistor (GFET), *presented at Bioengineering Day at UC San Diego*, San Diego, CA, 2023.

Darin Tsui. Design of a surgical navigation system via positional tracking of fiducial markers, *presented at the Conference on Micromechatronics for Information and Precision Equipment*, San Diego, CA, 2022.

Service

Reviewer: 2025 NeurIPS AI4Science Workshop, 2024 ICLR GEM Workshop.

Bioengineering Day Chair, Georgia Tech – Atlanta, GA

Jan 2025 - Present

- Co-organized Bioengineering Day, a symposium highlighting interdisciplinary research with over 100 attendees.
- Coordinated logistics, speaker outreach, and scheduling for faculty and alumni talks, award presentations, and a rapid-fire thesis competition.

Capstone Project Advisor, UC San Diego – San Diego, CA

Jul 2023 - Jun 2024

- Capstone project advisor for a bioengineering capstone project with Dr. Frank E. Talke.
- Mentored two undergraduates toward the design and development of a surgical navigation system with 3D tracking and augmented reality.
- Capstone project team won the *Teaching Assistant's Choice Award* at Bioengineering Day.

Founding Member, IEEE EMBS at UC San Diego – San Diego, CA

Apr 2023 - Jul 2023

- Co-founded the IEEE Engineering in Medicine and Biology Society (EMBS) at UC San Diego with Dr. Gert Cauwenberghs and other Ph.D. students to increase awareness of bioengineering topics and research.
- Recruited 5 graduate students to mentor undergraduate students interested in getting involved in research.

President, IEEE at UC San Diego – San Diego, CA

May 2022 - May 2023

- Oversaw operations for a 300+ student body. Partnered with the San Diego Supercomputing Center (SDSC) to co-found IEEE's Supercomputing Team.

- Hosted lab recruiting talks and technical seminars targeted towards first and second-years looking to get into research. Topics included research in artificial intelligence, robotics, and wearable sensors.
- Awarded the Outstanding IEEE Large Student Branch award for maintaining strong project activities and company relations during tenure.

Outreach Chair, IEEE at UC San Diego – San Diego, CA

May 2021 - May 2022

- Taught 60 elementary and middle school students programming basics at STEM Fair.
- Hosted a robotics workshop in collaboration with Robolink, a San Diego robotics company, where we invited 50 high school students from low-income communities to work with Robolink representatives on coding and designing their own aerial drones.

Project Team Lead, BMES at UC San Diego – San Diego, CA

Oct 2021 - Jun 2022

- Led team of 20 undergraduates in designing a telesurgical prosthetic hand at UC San Diego's Biomedical Engineering Society (BMES). Hand designs were formulated and constructed based on biomechanics principles.
- Held workshops in CAD, Arduino, and circuit design. Implemented robotic motion in Arduino and interfaced linear actuators.

Committee Member, BMES at UC San Diego – San Diego, CA

Sep 2019 - Jun 2023

- Committee member for BMES's Bioengineering Day committee, a day-long event celebrating all facets of bioengineering research and development, and Lab Expo committee, an undergraduate research symposium aiming to get students involved in research and academia.
- Recruited graduate students and professors to present research. Organized logistics and advertised events.

Mentoring

Kunal Talreja, ECE Undergraduate, Georgia Institute of Technology	2024 - 2025
Aryan Musharaf, Computer Science Undergraduate, Georgia Institute of Technology	2024 - 2025
Matthew Tam, Bioengineering Undergraduate, UC San Diego	2023 - 2024
Kirsten Ramos, Bioengineering Undergraduate, UC San Diego	2023 - 2024
Eric Lee, Bioengineering Undergraduate, UC San Diego	2023 - 2024
Capalina Melentyev, Bioengineering Undergraduate, UC San Diego	2022 - 2024
Rohan Kumar, Nanoengineering Undergraduate, UC San Diego	2022 - 2023